

AI ASSISTED CODING

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ASS 13.5

ask Description #1 (Refactoring – Removing Global Variables)

- Task: Use AI to eliminate unnecessary global variables from the code.

- Instructions:

- o Identify global variables used across functions.

- o Refactor the code to pass values using function parameters.

- o Improve modularity and testability.

- Sample Legacy Code:

```
rate = 0.1
```

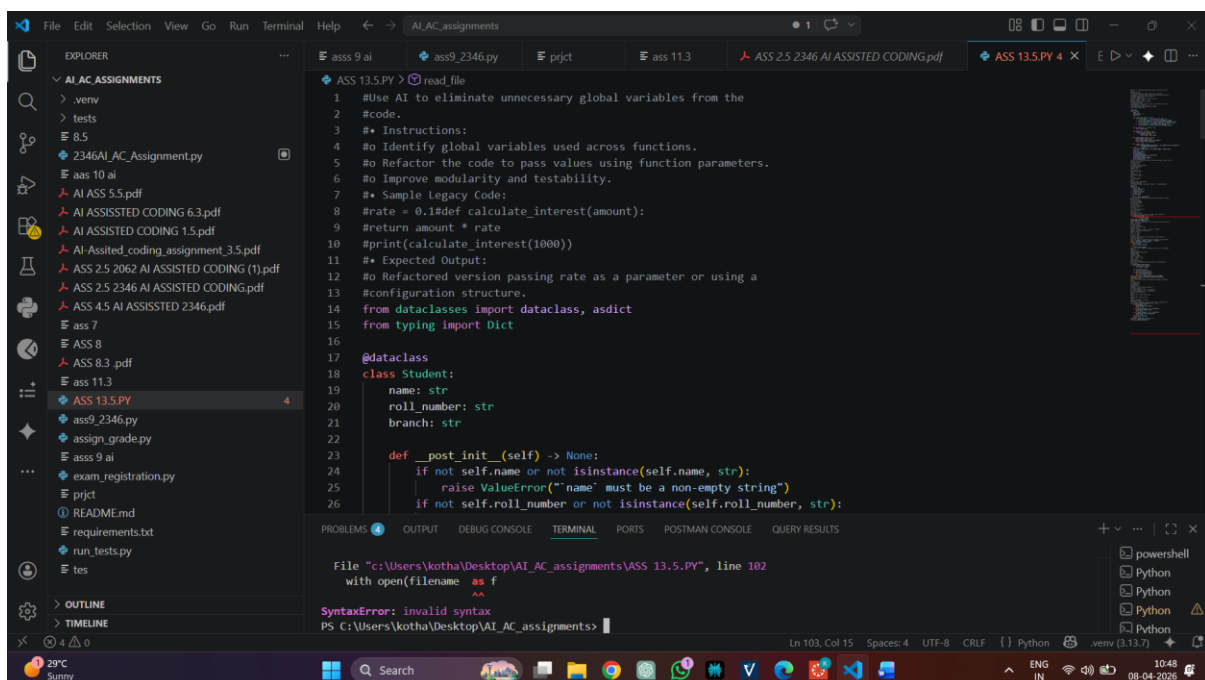
```
def calculate_interest(amount):
```

```
    return amount * rate
```

```
print(calculate_interest(1000))
```

- Expected Output:

- o Refactored version passing rate as a parameter or using a configuration structure.



Task Description #2 : (Refactoring Deeply Nested Conditionals)

- Task: Use AI to refactor deeply nested if–elif–else logic into a cleaner structure.

- Focus Areas:

- o Readability

- o Logical simplification

- o Maintainability

Legacy Code:

```
score = 78
```

```
if score >= 90:
```

```
    print("Excellent")
```

```
else:
```

```
    if score >= 75:
```

```
        print("Very Good")
```

```
    else:
```

```
        if score >= 60:
```

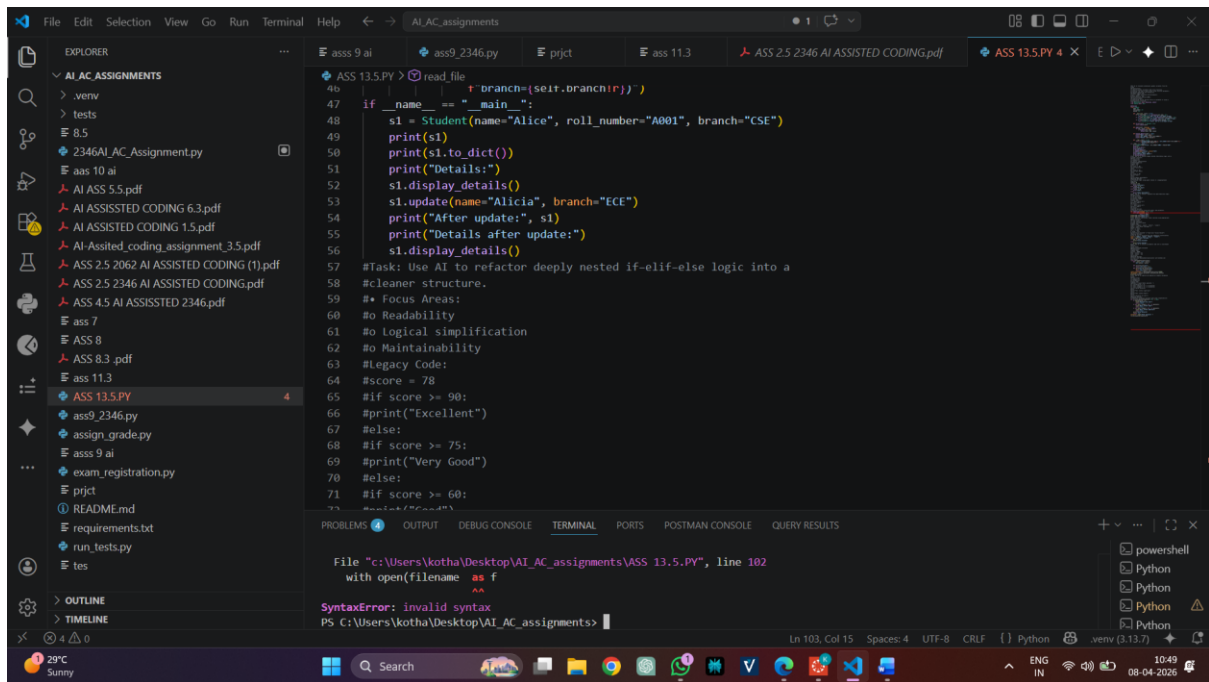
```
            print("Good")
```

```
        else:
```

```
            print("Needs Improvement")
```

Expected Outcome:

- o Flattened logic using guard clauses or a mapping-based approach



Task 3 (Refactoring Repeated File Handling Code)

- Task: Use AI to refactor repeated file open/read/close logic.

- Focus Areas:

- o DRY principle
- o Context managers
- o Function reuse

Legacy Code:

```
f = open("data1.txt")
```

```
print(f.read())
```

```
f.close()
```

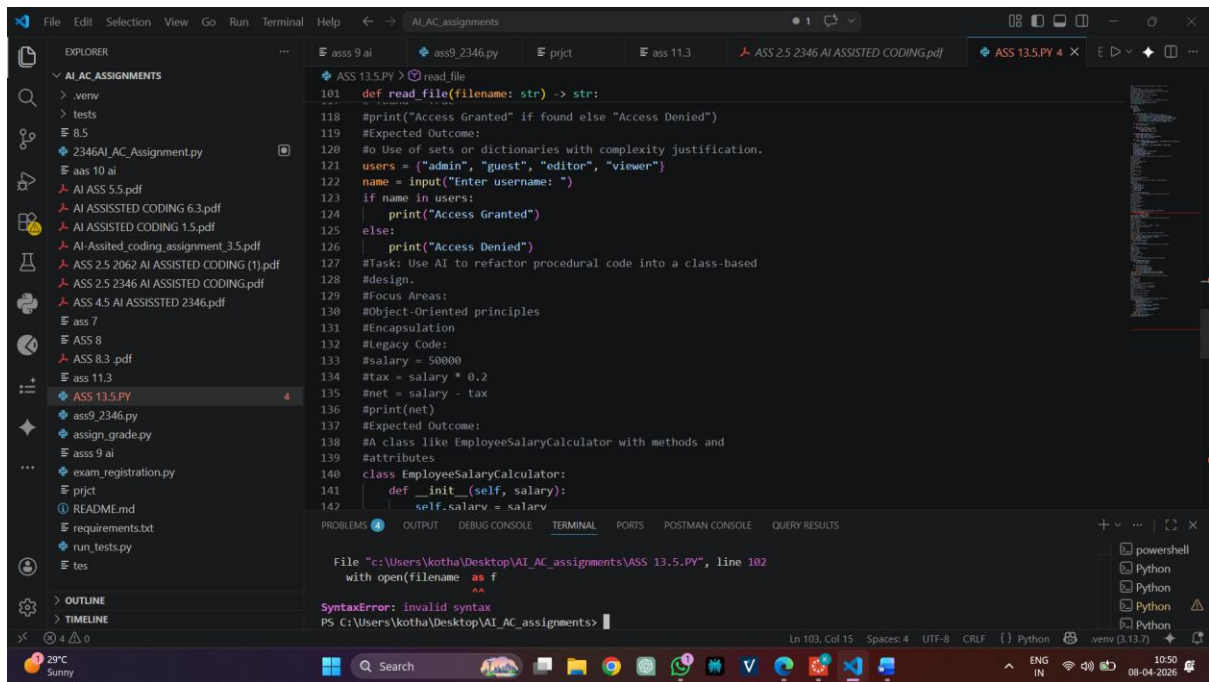
```
f = open("data2.txt")
```

```
print(f.read())
```

```
f.close()
```

Expected Outcome:

- o Reusable function using with open() and parameters.



(Refactoring Complex Input Validation Logic)

- Task: Use AI to simplify and modularize complex validation

rules.

- Focus Areas:

o Readability

o Testability

Legacy Code:

```
password = input("Enter password: ")
```

```
if len(password) >= 8:
```

```
if any(c.isdigit() for c in password):
```

```
if any(c.isupper() for c in password):
```

```
print("Valid Password")
```

```
else:
```

```
print("Must contain uppercase")
```

```
else:
```

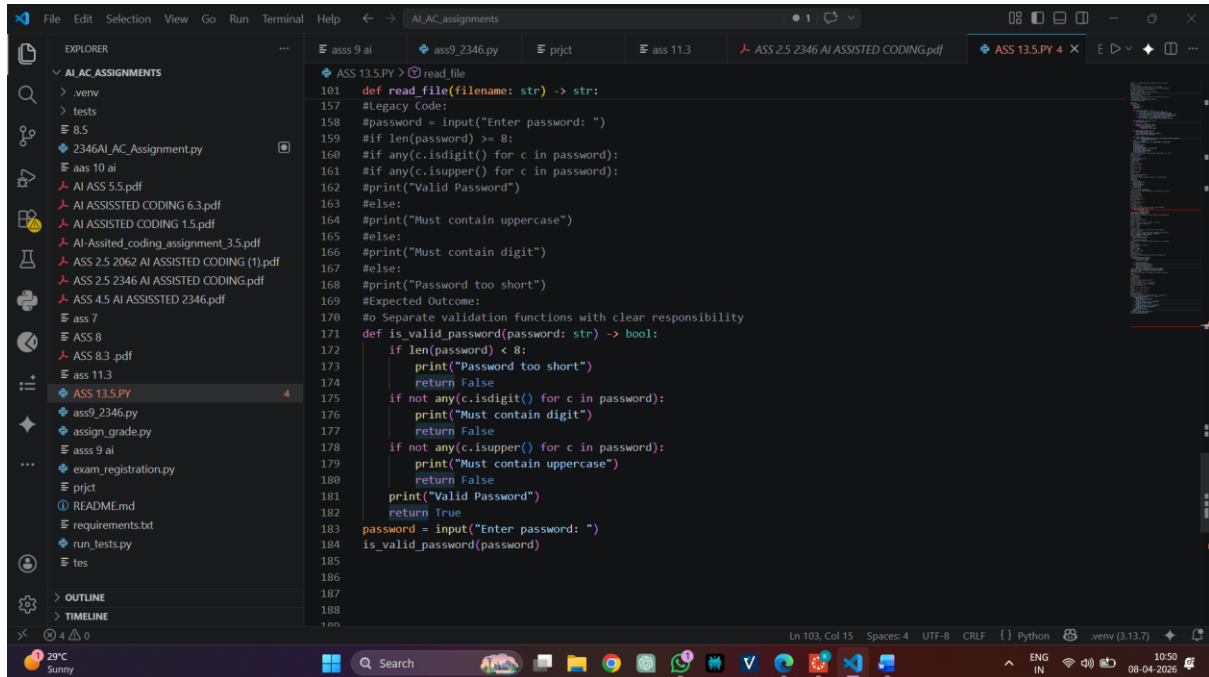
```
print("Must contain digit")
```

```
else:
```

```
print("Password too short")
```

Expected Outcome:

o Separate validation functions with clear responsibility



```
File Edit Selection View Go Run Terminal Help AI_AC_assignments
EXPLORER
  AI_AC_ASSIGNMENTS
    .venv
    tests
    8.5
    2346AI_AC_Assignment.py
    aas 10 ai
    AI ASS 5.5.pdf
    AI ASSISTED CODING 6.3.pdf
    AI ASSISTED CODING 1.5.pdf
    AI-Assited_coding_assignment_3.5.pdf
    ASS 2.5 2062 AI ASSISTED CODING (1).pdf
    ASS 2.5 2346 AI ASSISTED CODING.pdf
    ASS 4.5 AI ASSISTED 2346.pdf
    ass 7
    ASS 8
    ASS 8.3 .pdf
    ass 11.3
    ASS 13.5.PY 4
    ass9_2346.py
    assign_grade.py
    ass 9 ai
    exam_registration.py
    prjct
    README.md
    requirements.txt
    run_tests.py
    tes
  OUTLINE
  TIMELINE
  4 0 0
  23°C Sunny
  Search
  Ln 103, Col 15 Spaces: 4 UTF-8 CRLF Python .venv (3.13.7) 10:50 08-04-2026
```

```
ASS 13.5.PY > read file
101 def read_file(filename: str) -> str:
157 #Legacy Code:
158 #password = input("Enter password: ")
159 #if len(password) >= 8:
160 #if any(c.isdigit() for c in password):
161 #if any(c.isupper() for c in password):
162 #print("Valid Password")
163 #else:
164 #print("Must contain uppercase")
165 #else:
166 #print("Must contain digit")
167 #else:
168 #print("Password too short")
169 #Expected Outcome:
170 #o Separate validation functions with clear responsibility
171 def is_valid_password(password: str) -> bool:
172     if len(password) < 8:
173         print("Password too short")
174         return False
175     if not any(c.isdigit() for c in password):
176         print("Must contain digit")
177         return False
178     if not any(c.isupper() for c in password):
179         print("Must contain uppercase")
180         return False
181     print("Valid Password")
182     return True
183 password = input("Enter password: ")
184 is_valid_password(password)
185
186
187
188
189
```